

◆ THE COMPLETE LINE

```
app.run(host="0.0.0.0", port=5000)
```

Purpose:

Start the Flask web server.

Without this line:

Flask application exists

BUT

No server is running.

So nobody can access your API.

◆ BREAKDOWN

```
app
.
run(
    host="0.0.0.0",
    port=5000
)
```

Let's understand each part.

◆ WHAT IS app?



Earlier:

```
app = Flask(__name__)
```

created a Flask application **object**.

So:

```
app
```

contains your Flask application.

Think:

```
app
↓
Flask Application
```

◆ WHAT IS run() ?

```
run()
```

is a **method** of the Flask application object.

Just like:

```
lower() → String method
get() → Dictionary method
predict() → ML Model method
transform() → CountVectorizer method
```

Similarly:

```
run() → Flask Application method
```

◆ WHAT DOES run() DO?

```
app.run()
```

means:

```
Start listening for incoming HTTP requests.
```

Think:

```
Flask
↓
Waiting...
↓
Waiting...
↓
Waiting...
```

Until Spring Boot sends a request.

◆ WHAT IS host ?

```
host="0.0.0.0"
```

Host means:

```
Which network interface should Flask listen on?
```

◆ WHAT IS "0.0.0.0" ?

```
0.0.0.0
```

means:

Listen on all available network interfaces.

Think:

WiFi
Ethernet
Localhost

Flask accepts connections from all of them.

◆ WHY NOT localhost?

If you use:

`host="127.0.0.1"`



or

`host="localhost"`

then Flask only accepts connections from the same machine.

But:

`host="0.0.0.0"`



allows connections through any available network interface.

◆ WHAT IS port ?

`port=5000`

A port is like a door number for an application.

Think of a building:

Building
├─ Door 80
├─ Door 443
├─ Door 3306
└─ Door 5000

Your Flask application is waiting at:

Door 5000

◆ HOW DOES SPRING BOOT FIND FLASK?

In Spring Boot:

```
flask.api.url=http://localhost:5000/predict
```

Spring Boot sends requests to:

```
localhost
+
port 5000
+
/predict
```

So:

```
Spring Boot
  ↓
http://localhost:5000/predict
  ↓
Flask
```

◆ WHAT HAPPENS WHEN THIS LINE RUNS?

```
app.run(host="0.0.0.0", port=5000)
```

Step 1

```
Flask starts
```

↓

Step 2

```
Port 5000 opens
```

↓

Step 3

```
Flask waits for requests
```

↓

Step 4

```
Spring Boot sends POST /predict
```



IMP.

↓

Step 5

```
Flask receives request
```

↓

Step 6

```
predict() function executes
```

◆ WHY IS THIS AT THE BOTTOM?

Because Flask must first:

```
Load model
Load vectorizer
Create routes
Create functions
```

Only after everything is ready:

```
Start server
```

◆ NOTE FORMAT

```
app → Flask application object
run() → method used to start the Flask web server
host="0.0.0.0" → tells Flask to listen on all available network interfaces
port=5000 → tells Flask to run on port number 5000
👉 Use: start the Flask server and wait for incoming HTTP requests.
```

In your project:

```
Spring Boot
  ↓
http://localhost:5000/predict
  ↓
Flask Server (Port 5000)
  ↓
Spam Prediction
```